

Case report

Fatal acute methotrexate intoxication resulting from medication error in a rheumatoid arthritis patient: a case report

Xiangting Gao^{a,b,d}, Zhonghao Yu^{a,d}, Yuhao Yuan^a, Jiaxin Zhang^a, Longda Ma^c, Ying Dong^{a,e}, Yiwu Zhou^{a,*}^a Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, No. 13 Hangkong Road, Wuhan, 430030, People's Republic of China^b First Affiliated Hospital, Shihezi University/Department of Pathology and Key Laboratory for Xinjiang Endemic and Ethnic Diseases, Shihezi University School of Medicine, 107 North 2nd Road, Shihezi, People's Republic of China^c Postdoctoral Research Station in Basic Medicine, Hebei Medical University, Hebei, People's Republic of China

ARTICLE INFO

Keywords:

Methotrexate
Rheumatoid arthritis
Medication error
Adverse effects

ABSTRACT

Methotrexate (MTX) has long been a first-line anti-rheumatic agent due to its efficacy and cost-effectiveness. While generally safe at the recommended weekly dosage of 5–25 mg for rheumatoid arthritis, medication errors can lead to acute toxicity, severe adverse effects, and potentially fatal outcomes. We report a case of a 51-year-old male who experienced fatal pancytopenia following an MTX medication error during treatment for rheumatoid arthritis. His blood MTX concentration measured 0.04 $\mu\text{mol/L}$ 11 days after discontinuation. Post-mortem examination revealed MTX-related toxicity in multiple organs, including the liver and intestines. The spleen showed a prominent increase in plasma cells, including numerous Mott cells. Forensic analysis established the cause of death as acute MTX intoxication, directly stemming from a chain of medical negligence involving prescribing, verification, and administration failures. This case underscores the necessity of rigorous therapeutic drug monitoring, individualized dosing, and robust systemic safeguards to prevent iatrogenic fatalities, highlighting its critical medicolegal and educational implications for forensic and clinical practice.

1. Introduction

Rheumatoid arthritis (RA) is a chronic autoimmune disease characterized by joint inflammation, which significantly impacts patients' quality of life and functional status.¹ The management of RA primarily relies on pharmacological interventions, including drugs such as methotrexate (MTX), leflunomide, sulfasalazine, and hydroxychloroquine. Initially used at high doses in the treatment of leukemia, MTX was first evaluated for use in RA in 1951.² It received approval from the U.S. Food and Drug Administration for the treatment of RA in 1988.³ Due to its cost-effectiveness and significant clinical efficacy, low-dose MTX is now regarded as a cornerstone of RA therapy. However, its use is accompanied by a range of adverse effects, including gastrointestinal distress, rash, mucositis, myelosuppression, hepatotoxicity, and pulmonary fibrosis.⁴ These risks are heightened by medication errors, impaired

drug clearance, or inadequate monitoring adherence. Given its narrow therapeutic index, MTX requires precise dosing and rigorous surveillance to avoid serious, potentially life-threatening complications.

We report a case of fatal acute MTX intoxication resulting from a medication error in a patient with RA. Despite discontinuation of MTX 11 days prior to death, a residual serum concentration of 0.04 $\mu\text{mol/L}$ was detected. Postmortem examination revealed characteristic pathological findings, including hepatic necrosis, intestinal ulceration, and splenic plasmacytosis with Mott cells. This case underscores the importance of vigilant therapeutic drug monitoring and heightened clinical awareness of MTX-related toxicity.

2. Case history

A 51-year-old male with a two-year history of RA was admitted to a

* Corresponding author.

E-mail addresses: 1404467629@qq.com (X. Gao), 362736952@qq.com (Z. Yu), 1259970467@qq.com (Y. Yuan), 296034673@qq.com (J. Zhang), longdamafm@outlook.com (L. Ma), yingdong@hust.edu.cn (Y. Dong), zhouyiwu@hust.edu.cn (Y. Zhou).^d Equally contributed as First Author to this work.^e Equally contributed as Last Author to this work.<https://doi.org/10.1016/j.jflm.2026.103159>

Received 6 March 2025; Received in revised form 16 December 2025; Accepted 23 April 2026

Available online 24 April 2026

1752-928X/© 2026 Elsevier Ltd and Faculty of Forensic and Legal Medicine. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

city-level hospital in China due to a three-day exacerbation of poly-articular pain. Although oral ibuprofen had been prescribed for symptomatic relief at the time of diagnosis two years prior, he reported poor adherence to the medication regimen. Physical examination revealed swelling and restricted mobility in the metacarpophalangeal and interphalangeal joints of both hands. Laboratory investigations showed an elevated rheumatoid factor (RF) level of 112.00 IU/mL, confirming the RA diagnosis. The patient provided informed consent and commenced treatment with intravenous MTX at a dose of 100 mg daily for four consecutive days, alongside omeprazole.

Although his arthritic symptoms improved rapidly, the patient developed oral ulcers and pruritus of the head and neck on the fourth day. Attributing these to MTX, the physician discontinued the drug on day five and initiated supportive care, including hydration, leucovorin rescue, and management of anemia. By day seven, laboratory tests confirmed myelosuppression and hepatotoxicity, with alanine aminotransferase (ALT) and aspartate aminotransferase (AST) levels elevated to more than four times the upper limit of normal (Table 1). Supportive treatment was intensified with recombinant human granulocyte colony-stimulating factor (rhG-CSF) and hepatoprotective agents, which led to a partial recovery of liver function.

However, the myelosuppression persisted and progressively worsened. On day thirteen, the patient experienced severe diarrhea and a transient fever. By the morning of day fourteen, he presented with severe pancytopenia, fever, and coagulopathy. His condition rapidly deteriorated that evening, culminating in dyspnea, refractory hypotension, and coma. The patient died on the fifteenth day of hospitalization (Fig. 1).

3. Autopsy findings

3.1. Gross examination

A forensic autopsy was conducted seven days post-mortem. External examination revealed no significant injuries. Internal examination of the thorax showed a midsternal fracture, likely related to resuscitation attempts. Notable findings included severe pulmonary edema and congestion, with pleural adhesions between the lungs and chest wall (right lung: 1407 g; left lung: 958 g). The heart weighed 500 g and demonstrated left ventricular hypertrophy, with a wall thickness of 1.5 cm. Coronary artery examination revealed 50% occlusion in both the left and right arteries.

3.2. Histological findings

Histopathological examination revealed multi-organ damage. Lung tissues exhibited severe congestion, edema, extensive hemorrhage, and alveolar wall fibrosis, accompanied by hyaline membrane formation, mild bronchiolitis, and abundant intra-alveolar macrophages (Fig. 2A and B). The liver showed mild steatosis with scattered hepatocyte necrosis (Fig. 2C). Myocardial tissue demonstrated cardiomyocyte hypertrophy and focal replacement fibrosis. The interstitium of the left ventricular wall contained numerous Aschoff cells (Fig. 2D). Renal analysis revealed extensive necrosis of the renal tubules (Fig. 2E). In the spleen, lymphocyte necrosis was observed alongside an expansion of plasma cells, many of which were Mott cells containing Russell bodies

Table 1
Key laboratory test results following MTX administration.

Variables	Day 1	Day 7	Day 8	Day 10	Day 14	Reference Range
White Blood Cells ($\times 10^9/L$)	5.36	1.46	2.73	1.67	0.60	3.5-9.5
Hemoglobin (g/L)	106	99	98	103	80	115-150
Platelets ($\times 10^9/L$)	217	157	149	103	18	125-350



Fig. 1. Timeline of the clinical course following MTX administration. Days 1-4: Intravenous MTX (100 mg daily), Day 4: Onset of adverse effects (oral ulcers, skin itching), Day 5: MTX discontinuation and initiation of supportive care, Day 7: Confirmation of myelosuppression and hepatotoxicity, Day 13: Severe diarrhea and transient fever, Day 14: Severe pancytopenia, coagulopathy, and rapid deterioration to shock and coma, Day 15: Patient death.

(Fig. 2F), the central arteries also exhibited hyaline degeneration. Brain tissue displayed edema with evidence of neuronophagia. The mucosa of the small intestine and colon showed a prominent lymphocytic infiltrate.

4. Toxicological analysis

Peripheral blood samples were collected to assess the MTX blood concentration. The analysis was performed using LC-MS/MS. The standard curve was determined using the least squares method, resulting in equation $Y = 912.54X + 169.211$, $r = 0.995$ (Fig. 3A). MTX was detected in the blood sample, and the final MTX blood concentration was determined to be 18.6 ng/mL (0.04 $\mu\text{mol/L}$), as depicted in Fig. 3B.

5. Medicolegal investigation

Based on the comprehensive investigation, autopsy findings, and toxicological analysis, it was conclusively determined that the death resulted from acute MTX intoxication. The court adjudicated that the healthcare institution and involved medical practitioners were liable for medical negligence and held legally responsible for the incident.

6. Discussion

MTX is widely used for RA and other inflammatory diseases. It acts by inhibiting dihydrofolate reductase, thus impairing DNA synthesis and immune cell proliferation.⁵ Despite the relative safety of low-dose MTX in RA, adverse effects such as infections, hypoalbuminemia, pneumonitis, hepatotoxicity, renal impairment, mucositis, and pancytopenia.⁶ A pooled analysis underscores the prevalence of adverse events, with over 70% of patients in prospective studies experiencing at least one.⁷ MTX toxicity is linked to dosage, treatment duration, and individual risk factors, and dosage errors can result in severe or fatal consequences.

In this case, the patient developed multiple clinical manifestations following MTX administration, including oral ulcers, elevated liver enzymes, pruritus, and diarrhea. The subsequent onset of severe pancytopenia significantly increased the risk of infection and septic shock, a condition with a reported mortality rate as high as 44%.⁸ Postmortem examination revealed signs indicative of shock, including severe pulmonary edema and pulmonary hemorrhage, which is consistent with previous reports of acute MTX intoxication.⁹ While Aschoff cells observed in the myocardial interstitium suggested rheumatic myocarditis, the lesion was mild and not considered a direct cause of acute heart failure. Based on the patient's history of enteritis and bronchitis,

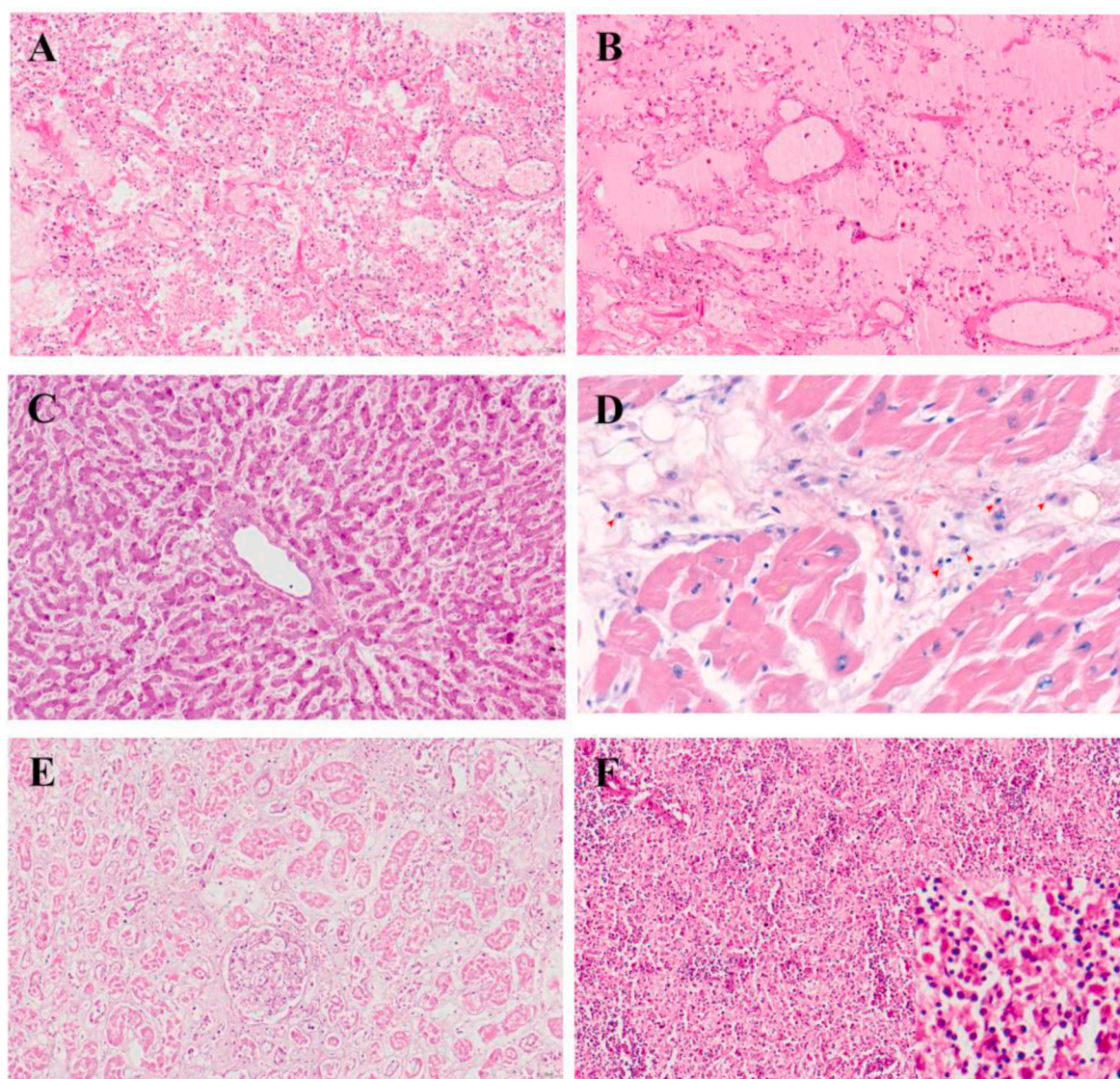


Fig. 2. Pathological findings in multiple organs. (A) Pulmonary hyaline membranes and extensive hemorrhage, (B) Mild bronchiolitis and abundant intra-alveolar macrophages, (C) Hepatic steatosis with scattered hepatocyte necrosis, (D) Scattered Aschoff cells (arrowheads) in the left ventricular myocardium, (E) Extensive renal tubular necrosis, (F) Splenic lymphocyte necrosis and plasmacytosis with Mott cells containing Russell bodies.

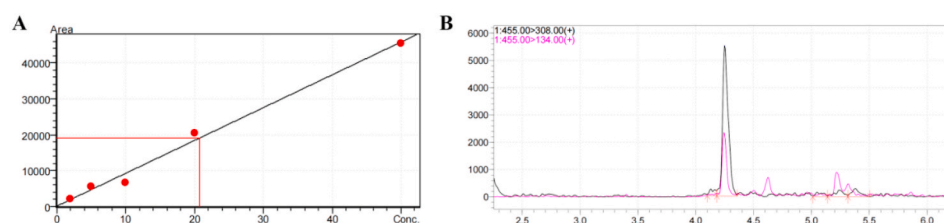


Fig. 3. The chromatogram of MTX from the blood sample. (A) The standard curve of MTX, (B) The blood concentration of MTX.

compounded by the documented severe pancytopenia and fever, the cause of death was ultimately attributed to septic shock. This fatal sequence of events was precipitated by pancytopenia secondary to acute MTX intoxication.

Common causes of MTX toxicity include patient-related medication errors, such as accidental daily intake or overdose, and healthcare system errors, including electronic prescription mistakes by physicians or dispensing inaccuracies by pharmacists.^{10–15} Literature reports indicate that erroneous MTX dosage is associated with a mortality rate ranging

from 5.2% to 24%.¹⁶ The recommended dosage for RA is 5 to 25 mg per week, administered orally or subcutaneously.¹⁷ In the present case, however, the patient received 100 mg intravenously daily for four consecutive days—a dose exceeding the recommended maximum by more than 15-fold. Critically, the route of administration compounded the error: intravenous delivery resulted in immediate and complete bioavailability, bypassing the variable absorption associated with oral or subcutaneous routes. This led to rapidly sustained peak serum concentrations, creating a pharmacokinetic assault that fundamentally

distinguished this incident from a mere dosage miscalculation and directly contributed to its fulminant course.

Furthermore, an inappropriate drug combination involving omeprazole was identified. Co-administration of omeprazole may result in a pharmacokinetic interaction that elevates plasma MTX concentrations, thereby increasing toxicity risk.^{18,19} These cumulative medication errors directly culminated in the observed acute MTX intoxication. In this instance, the primary error occurred at the prescribing stage, where a junior physician erroneously prescribed a chemotherapeutic MTX dosage for RA. This mistake was subsequently overlooked by both the dispensing pharmacist and the administering nurse, who followed the prescribed instructions, constituting a chain of medical negligence. The court adjudicated that this negligence substantially reduced the patient's chance of survival. Consequently, the responsible healthcare institution and practitioners were found liable for medical negligence, with corresponding financial compensation mandated.

A notable pathological finding in this case was the increase of plasma cells and the presence of numerous Mott cells in the spleen—a phenomenon rarely reported in the context of MTX treatment for RA. Histologically, Mott cells are a variant of plasma cells characterized by an eccentric nucleus and a cytoplasm packed with spherical, eosinophilic Russell bodies, which are dilated endoplasmic reticulum cisternae containing condensed, non-secreted immunoglobulins. While typically associated with conditions such as multiple myeloma, reactive plasmacytosis, or plasma cell neoplasms, their appearance here may be linked to the administration of recombinant human granulocyte colony-stimulating factor (rhG-CSF). rhG-CSF is known to enhance interleukin-6 secretion by monocytes and macrophages, a cytokine that potentially drives plasma cell differentiation and expansion.²⁰ The emergence of Mott cells, therefore, provides direct histomorphological evidence of a pronounced, rhG-CSF-induced B-cell dysregulation, representing an aberrant stage of plasma cell activation within the splenic tissue. Regarding hematological toxicity, a cohort study of 70 patients with MTX-associated pancytopenia revealed that 17% failed to recover from myelosuppression and succumbed to the complication.²¹ This is consistent with established evidence that high-dose MTX can induce significant bone marrow suppression.²² In the present case, the patient developed severe pancytopenia. Unfortunately, a bone marrow examination was not performed, precluding a definitive assessment of the marrow's status at the time of death.

Therapeutic drug monitoring of serum MTX is essential to mitigate the risk of adverse effects during treatment.²³ The plasma half-life of low-dose MTX is approximately 8–10 hours and the drug is predominantly eliminated via renal excretion.²⁴ In the present case, a blood sample collected 11 days after the last MTX administration showed a concentration of 0.04 $\mu\text{mol/L}$, indicating near-complete elimination. However, MTX may occasionally exhibit delayed clearance, with sustained detectable concentrations several days after discontinuation.²⁵ Therefore, even after several days of withdrawal, toxicological analysis may still reveal elevated MTX levels. It is critical to note that a normal MTX concentration does not rule out the possibility of MTX intoxication, as tissue damage may persist despite clearance from the systemic circulation.

In summary, MTX intoxication exhibits distinct clinical and laboratory abnormalities that serve as vital evidence for forensic identification. A comprehensive histological assessment of multiple organs—ideally including bone marrow—is essential for forensic pathologists to conclusively diagnose fatal acute MTX intoxication. Beyond diagnosis, this case serves as an instructive paradigm, demonstrating the critical interface between forensic pathology and clinical risk management and providing concrete evidence to advocate for systemic reforms aimed at preventing future recurrence.

Informed consent

Written informed consent was obtained from the family members of

the patient for publication of the case details.

Funding

The authors received no financial support for the research, authorship, and publication of this article.

Conflict of interest statement

All authors have agreed and declared that there was no conflict of interest.

Acknowledgements

No.

References

- Di Matteo A, Bathon JM, Emery P. Rheumatoid arthritis. *Lancet*. 2023;402(10416): 2019–2033. [https://doi.org/10.1016/S0140-6736\(23\)01525-8](https://doi.org/10.1016/S0140-6736(23)01525-8).
- Brown PM, Pratt AG, Isaacs JD. Mechanism of action of methotrexate in rheumatoid arthritis, and the search for biomarkers. *Nat Rev Rheumatol*. 2016;12(12):731–742. <https://doi.org/10.1038/nrrheum.2016.175>.
- Braun J, Rau R. An update on methotrexate. *Curr Opin Rheumatol*. 2009;21(3): 216–223. <https://doi.org/10.1097/BOR.0b013e328329c79d>.
- Solomon DH, Glynn RJ, Karlson EW, et al. Adverse effects of low-dose methotrexate: a randomized trial. *Ann Intern Med*. 2020;172(6):369–380. <https://doi.org/10.7326/M19-3369>.
- Zhao Z, Hua Z, Luo X, et al. Application and pharmacological mechanism of methotrexate in rheumatoid arthritis. *Biomed Pharmacother*. 2022;150, 113074. <https://doi.org/10.1016/j.biopha.2022.113074>.
- Pivovarov K, Zipursky JS. Low-dose methotrexate toxicity. *Can Med Assoc J*. 2019; 191(15), E423. <https://doi.org/10.1503/cmaj.181054>.
- Salliot C, van der Heijde D. Long-term safety of methotrexate monotherapy in patients with rheumatoid arthritis: a systematic literature research. *Ann Rheum Dis*. 2009;68(7):1100–1104. <https://doi.org/10.1136/ard.2008.093690>.
- Kuitunen T, Malmstrom J, Palva E, et al. Pancytopenia induced by low-dose methotrexate: a study of the cases reported to the Finnish Adverse Drug Reaction Register from 1991 to 1999. *Scand J Rheumatol*. 2005;34(3):238–241. <https://doi.org/10.1080/03009740510018570>.
- Moisa A, Fritz P, Benz D, et al. Iatrogenically-related, fatal methotrexate intoxication: a series of four cases. *Forensic Sci Int*. 2006;156(2–3):154–157. <https://doi.org/10.1016/j.forsciint.2004.12.031>.
- Cairns R, Brown JA, Lynch AM, et al. A decade of Australian methotrexate dosing errors. *Med J Aust*. 2016 Jun 6;204(10):384. <https://doi.org/10.5694/mja15.01242>.
- Thompson JA, Love JS, Hendrickson RG. Methotrexate toxicity from unintentional dosing errors: calls to a poison Center and death descriptions. *J Am Board Fam Med*. 2021 Nov-Dec;34(6):1246–1248. <https://doi.org/10.3122/jabfm.2021.06.210120>.
- Singh A, Handa AC. Medication Error- A case report of misadventure with methotrexate. *JNMA J Nepal Med Assoc*. 2018 Mar-Apr;56(211):711–715.
- Ferner RE. Medication errors that have led to manslaughter charges. *BMJ*. 2000 Nov 11;321(7270):1212–1216. <https://doi.org/10.1136/bmj.321.7270.1212>.
- Oyebode F. Clinical errors and medical negligence. *Med Princ Pract*. 2013;22(4): 323–333. <https://doi.org/10.1159/000346296>.
- Jesiah N, Siva Y, Kt S. Fatal neutropenia sepsis following acute methotrexate toxicity. *Cureus*. 2025 Jan 21;17(1), e77754. <https://doi.org/10.7759/cureus.77754>.
- Szpot P, Wacheiko O, Zawadzki M. Toxicological aspects of methotrexate intoxication: concentrations in postmortem biological samples and autopsy findings. *Toxics*. 2022 Sep 29;10(10):572. <https://doi.org/10.3390/toxics10100572>.
- Smolen JS, Landewe R, Bergstra SA, et al. EULAR recommendations for the management of rheumatoid arthritis with synthetic and biological disease-modifying antirheumatic drugs: 2022 update. *Ann Rheum Dis*. 2023;82(1):3–18. <https://doi.org/10.1136/ard-2022-223356>.
- Narumi K, Sato Y, Kobayashi M, et al. Effects of proton pump inhibitors and famotidine on elimination of plasma methotrexate: evaluation of drug-drug interactions mediated by organic anion transporter 3. *Biopharm Drug Dispos*. 2017;38 (9):501–508. <https://doi.org/10.1002/bdd.2091>.
- Ueda H, Narumi K, Sato Y, et al. Evaluation of possible pharmacokinetic interaction between methotrexate and proton pump inhibitors in rats. *Pharmacol Rep*. 2020;72 (5):1426–1432. <https://doi.org/10.1007/s43440-020-00130-1>.
- Stonedahl S, Leser JS, Clarke P, et al. Treatment with granulocyte-macrophage colony-stimulating factor reduces viral titers in the brains of West Nile virus-infected mice and improves survival. *J Virol*. 2023;97(3), e180522. <https://doi.org/10.1128/jvi.01805-22>.
- Gutierrez-Urena S, Molina JF, Garcia CO, et al. Pancytopenia secondary to methotrexate therapy in rheumatoid arthritis. *Arthritis Rheum*. 1996;39(2):272–276. <https://doi.org/10.1002/art.1780390214>.
- Hamed KM, Dighiriri IM, Baomar AF, et al. Overview of methotrexate toxicity: a comprehensive literature review. *Cureus J Med Science*. 2022;14(9), e29518. <https://doi.org/10.7759/cureus.29518>.

23. Howard SC, McCormick J, Pui CH, et al. Preventing and managing toxicities of high-dose methotrexate. *Oncologist*. 2016;21(12):1471–1482. <https://doi.org/10.1634/theoncologist.2015-0164>.
24. Ouellette S, Shah R, Razi S, et al. Fatal low-dose methotrexate toxicity: a case report and literature review. *Dermatol Ther*. 2022;35(12), e15945. <https://doi.org/10.1111/dth.15945>.
25. Bielack S, Fox CP, Hoang-Xuan K, et al. A Delphi study to determine the epidemiology and clinical management of patients treated with HDMTX who develop methotrexate (MTX) delayed elimination in France, Germany, Italy, and the UK. *Health Sci Rep*. 2024;7(1), e1749. <https://doi.org/10.1002/hsr2.1749>.